



Implementation Plan & Roadmap

Phase 1: Infrastructure & DB Setup

1. Initialize Python FastAPI backend and Next.js frontend repositories.
2. Set up Supabase PostgreSQL with `pgvector` enabled.
3. Implement secure Auth endpoints, strict CORS, and rate-limiting (`slowapi`).

Phase 2: Web API & ML Integration

1. Integrate the pre-trained XGBoost model and SHAP explainer into a FastAPI endpoint.
2. Deploy the LangGraph pipeline (`insights_agent.py`) inside FastAPI to process the ML payload and return structured Pydantic JSON.
3. Create CRUD endpoints for `mood_logs` and `ai_insights_cache` .

Phase 3: RAG & Chatbot Pipeline

1. Set up local or Hugging Face embedding endpoints for text chunking.
2. Build the Aura chat WebSocket/REST endpoints integrating the Vector DB retrieval step.
3. Implement the daily cron job for RLHF nudges, logging to `user_nudge_feedback` .

Phase 4: Frontend "Organic Calm" UI

1. Configure Tailwind CSS with the Forest Green/Sage/Cream palette.
2. Build the Hero Dashboard with overlapping pill-shaped action bars.
3. Connect the dashboard to the LangGraph endpoint to render the sliding deep-insight cards.
4. Build the Aura Chat interface.

Phase 5: Security Audit & Deployment

1. Audit all endpoints for SQLi and prompt injection vulnerabilities.
2. Deploy Backend to a GPU-enabled cloud (e.g., RunPod, AWS, or Hugging Face Dedicated).
3. Deploy Frontend to Vercel over HTTPS.